

Anthony Bardou

RESEARCH SCIENTIST – MACHINE LEARNING, OPTIMIZATION & AI

☎ +33 669 302 564 | ✉ anthony.bardou@epfl.ch | 🏠 abardou.github.io | 📱 abardou | 🌐 anthony-bardou-437b96170

Research scientist specializing in machine learning and optimization for decision-making under uncertainty. My work combines theoretical analysis with practical algorithm design, with a focus on Bayesian optimization, online learning, and high-dimensional black-box problems in dynamic and decentralized settings. I also explore efficient evaluation methods for language models and develop open-source research software. My work has been published at ICLR, NeurIPS, and in leading machine-learning and networking journals, and has received best paper and best PhD thesis awards.

Core Expertise

ML & AI	Bayesian Optimization, Online Learning, Bandits, LLM Evaluation, Probabilistic Models
Optimization	Zeroth-Order, Stochastic, High-Dimensional, Dynamic & Decentralized Optimization
Engineering	Python, PyTorch, BoTorch, NumPy/SciPy, C/C++, Git, Linux
Languages	French (native), English (full professional proficiency), Spanish and Romanian (basic)

Experience

École Polytechnique Fédérale de Lausanne

Lausanne, CH

POSTDOCTORAL RESEARCHER – MACHINE LEARNING AND OPTIMIZATION

Oct. 2023 - Present

- Develop scalable Bayesian optimization methods for noisy, high-dimensional, dynamic, and symmetry-rich black-box problems, combining theoretical guarantees with empirical evaluation.
- Published at ICLR and NeurIPS; released reusable PyTorch/BoTorch implementations and contributed to a benchmark for diffusion language models.
- Collaborate with researchers across machine learning, optimization, and networked systems; supervised nine Bachelor's and Master's research projects.

Keywords: Bayesian Optimization, Online Learning, Dynamic Systems, PyTorch, BoTorch

École Normale Supérieure de Lyon

Lyon, FR

RESEARCH INTERN – ONLINE LEARNING FOR WIRELESS SYSTEMS

Apr. 2020 - Sep. 2020

- Designed multi-armed-bandit methods for adaptive spatial reuse in dense Wi-Fi networks and evaluated them in a custom network simulator.
- Established the research direction later developed into an award-winning PhD thesis and multiple peer-reviewed publications.

Keywords: Online Learning, Multi-Armed Bandit, Spatial Reuse

Université Claude Bernard Lyon I

Lyon, FR

RESEARCH INTERN – NATURAL LANGUAGE PROCESSING

Dec. 2018 - Jun. 2019

- Developed attention and representation-learning methods to classify citation intent in scientific documents, improving F1 by 23 points on internal benchmarks.

Keywords: Natural Language Processing, Embedding, Attention, Deep Learning, PyTorch

Université de Perpignan Via Domitia

Perpignan, FR

UNDERGRADUATE RESEARCH INTERN – OPTIMIZATION AND COMPUTER VISION

Apr. 2018 - Jun. 2018

- Analyzed the robustness of a non-invasive computer-vision system for respiratory-volume measurement and reduced its computational complexity using combinatorial optimization.

Keywords: Combinatorial Optimization, Robustness Analysis, Image Processing

Education

2020–23	PhD in Machine Learning and Wireless Networks , École Normale Supérieure de Lyon — <i>Online Learning for the Black-Box Optimization of Wireless Networks</i> ; Best PhD Thesis Award	Lyon, FR
2018–20	M.S. in Data Science , Université Claude Bernard Lyon I — ranked 1st of 47, highest honors	Lyon, FR
2015–18	B.S. in Computer Science and Mathematics , Université de Perpignan Via Domitia — ranked 1st of 16, highest honors	Perpignan, FR

Honors & Awards

INTERNATIONAL

2025	Top Reviewer , 39th Annual Conference on Neural Information Processing Systems	San Diego, US
2024	Top Reviewer , 38th Annual Conference on Neural Information Processing Systems	Vancouver, CA
2022	Best Paper Award , 25th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems	Montreal, CA

NATIONAL

2025	Best Ph.D. Thesis , GDR Réseaux et Systèmes Distribués / Association ACM SIGOPS France	Toulouse, FR
2022	Best Student Paper Award , 24ème Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications	Paris, FR

Open-Source Research Software

speedrun-dlm: Efficient benchmarking for diffusion language models	Jul. 2026	agonon/speedrun-dlm
max-kernel: Symmetry-aware kernels for Bayesian optimization	Feb. 2026	abardou/max-kernel
W-DBO: Wasserstein Distance-Based Dynamic Bayesian Optimization	May 2024	WDBO-ALGORITHM/wdbo_algo
DuMBO: High-Dimensional, No-Regret Bayesian Optimization	Feb. 2023	abardou/dumbo

Schools & Workshops

Swiss Optimization Symposium (SOS'26)	Aug. 2026	Ascona, CH
Spring School of Theoretical Computer Science on Machine Learning (EPIT'22)	May 2022	Luminy, FR

Talks

Jul. 2026	Invited Speaker , L2S (Paris Saclay University, CentraleSupélec)	Paris, FR
Sep. 2025	Invited Speaker , ICube Lab (Strasbourg University)	Strasbourg, FR
May 2025	Invited Speaker , Secondmind Research Seminar	Cambridge, UK
Mar. 2025	Invited Speaker , Journées GDR Réseaux et Systèmes Distribués	Toulouse, FR
Jun. 2024	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
May 2023	Invited Speaker , INDY Lab (EPFL)	Lausanne, CH
Oct. 2022	Oral Presentation , MSWiM'22	Montreal, CA
Jul. 2022	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
Jun. 2022	Oral Presentation , AlgoTel'22	Paris, FR
Feb. 2022	Oral Presentation , ROADEF'22	Lyon, FR
Dec. 2021	Invited Speaker , INRIA Ockham (ENS Lyon)	Lyon, FR
May 2021	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
Nov. 2020	Invited Speaker , Performance Evaluation for Networks	Lyon, FR
Jul. 2018	Invited Speaker , LIRMM (Perpignan University)	Perpignan, FR

External Accolades

Prix de thèse du GDR RSD : des travaux sur l'optimisation récompensés	Jan. 2025	INS2I (CNRS)
Lauréat du prix de thèse GDR RSD - Édition 2024	Sep. 2024	GDR RSD (CNRS)
L'optimisation bayésienne au service des performances du WiFi	Apr. 2023	INS2I (CNRS)

Leadership, Teaching & Service

PEER REVIEW (YEAR FIRST REVIEWED)

2025	International Conference on Learning Representations , International Conference
2025	Scientific Reports , Peer-Reviewed Journal (Nature)
2025	Transactions on Machine Learning Research , Peer-Reviewed Journal
2025	International Conference on Machine Learning , International Conference
2024	Computer Networks , Peer-Reviewed Journal (Elsevier)
2023	Neural Information Processing Systems , International Conference
2023	Computer Communications , Peer-Reviewed Journal (Elsevier)
2023	International Symposium on Information Theory , International Conference (IEEE)

TEACHING

Fall 2024–26	Stochastic Modeling , 30 hours @ EPFL (Bachelor)	Lausanne, CH
Spring 2023	Preparation to the <i>agrégation</i> of Computer Science , 32 hours @ ENS Lyon (Master)	Lyon, FR
Spring 2022	Numerical Methods , 15 hours @ UCBL (Bachelor)	Lyon, FR
Spring 2022	Optimization , 15 hours @ UCBL (Bachelor)	Lyon, FR
Fall 2021	Introduction to Artificial Intelligence , 12 hours @ UCBL (Master)	Lyon, FR
Fall 2021	Optimization and Operations Research , 18 hours @ UCBL (Master)	Lyon, FR

SUPERVISION AND MENTORING

Spring 2026	Marin Philippe , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Robin Jaccard , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Adrien Deschenaux , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2025	Aryan Ahadinia , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Antoine Merel , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Nahuel Plomb Veuve , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2024	Emanuele Mengoli , Master Thesis @ EPFL (from École Polytechnique)	Lausanne, CH
Spring 2024	Giovanni Ranieri , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2023	Hugo Martel , Bachelor Research Internship @ ENS Lyon	Lyon, FR

JURIES AND COMMITTEES

Jul. 2026	External Expert , Master Thesis Defense of David Kalajdzic	EDIC, EPFL (CH)
Dec. 2025	Committee Member , Evaluation of prospective PhD students	EDIC, EPFL (CH)
May 2024	External Expert , Evaluation of prospective PhD students	INRIA Lyon (FR)

Publications (10 peer-reviewed papers)

INTERNATIONAL CONFERENCES

ANTHONY BARDOU, ANTOINE GONON, ARYAN AHADINIA, PATRICK THIRAN. Symmetry-Aware Bayesian Optimization via Max Kernels. *Proc. ICLR'26*. *Rio de Janeiro, BR*. Apr. 2026

ANTHONY BARDOU, PATRICK THIRAN, GIOVANNI RANIERI. This Too Shall Pass: Removing Stale Observations in Dynamic Bayesian Optimization. *Proc. NeurIPS'24*. *Vancouver, CA*. Dec. 2024

ANTHONY BARDOU, PATRICK THIRAN, THOMAS BEGIN. Relaxing the Additivity Constraints in Decentralized No-Regret High-Dimensional Bayesian Optimization. *Proc. ICLR'24*. *Vienna, AT*. May 2024.

ANTHONY BARDOU, THOMAS BEGIN. INSPIRE: Distributed Bayesian Optimization for ImproviNg SPatIal REuse in Dense WLANs. *Proc. MSWiM'22*. *Montréal, CA*. Oct. 2022. 10.1145/3551659.3559050. **BEST PAPER AWARD**.

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Improving the Spatial Reuse in IEEE 802.11ax WLANs: A Multi-Armed Bandit Approach. *Proc. MSWiM'21*. *Alicante, ES*. Nov. 2021. 10.1145/3479239.3485715.

PEER-REVIEWED JOURNALS

ANTHONY BARDOU, JEAN-MARIE GORCE, THOMAS BEGIN. Assessing the Performance of NOMA in a Multi-Cell Context: A General Evaluation Framework. *IEEE Transactions on Wireless Communications*. Jul. 2025.

SAMIR SI-MOHAMMED, **ANTHONY BARDOU**, THOMAS BEGIN, ISABELLE GUÉRIN-LASSOUS, PASCALE VICAT-BLANC. NS+ NDT: Smart Integration of Network Simulation in Network Digital Twin, Application to IoT Networks. *Future Generation Computer Systems*. Mar. 2024.

ANTHONY BARDOU, THOMAS BEGIN. Analysis of a Decentralized Bayesian Optimization Algorithm for Improving Spatial Reuse in Dense WLANs. *Computer Communications, ACM MSWiM 2022 Special Issue*. Jun. 2023. 10.1016/j.comcom.2023.06.004

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Mitigating Starvation in Dense WLANs: A Multi-Armed Bandit Solution. *Ad Hoc Networks, Vol. 138*. Jan. 2023. 10.1016/j.adhoc.2022.103015

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Analysis of a Multi-Armed Bandit Solution for Improving the Spatial Reuse of New Generation WLANs. *Computer Communications, ACM MSWiM 2021 Special Issue*. Sep. 2022. 10.1016/j.comcom.2022.07.015

PH.D. THESIS

ANTHONY BARDOU. Online Learning for the Black-Box Optimization of Wireless Networks. *ENS Lyon. Lyon, FR*. Sep. 2023. **GDR RSD/ASF BEST PH.D. THESIS AWARD.**